Logistic Regression: From Probability to Evaluation

Modeling, fitting, and evaluating binary outcomes

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1.

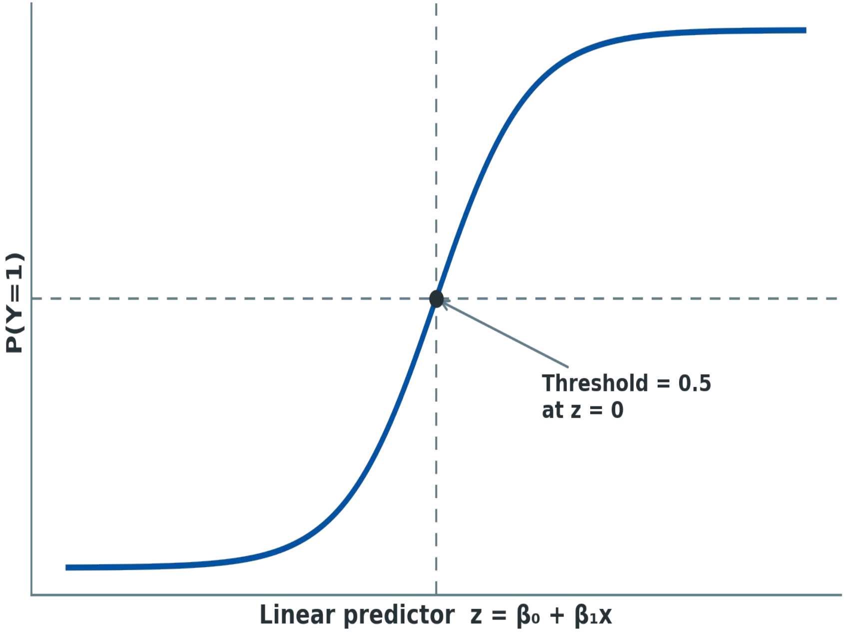
Part 1

Model Foundations

From Linear Predictor to Probability

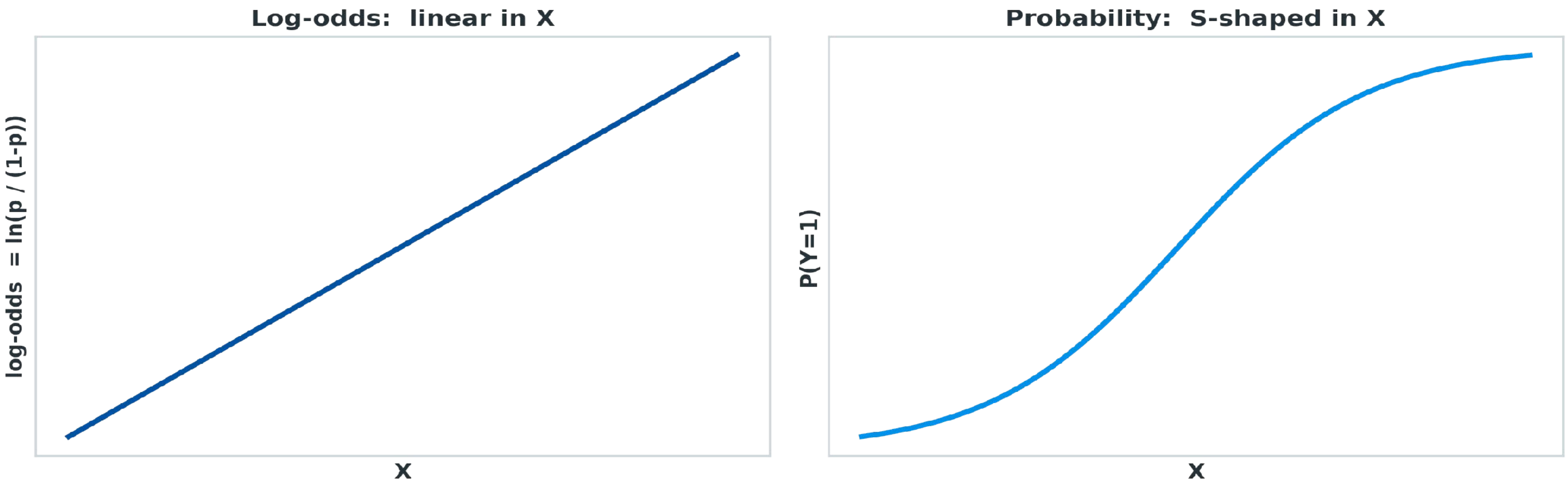
- A linear predictor $z = \beta_0 + \beta_1 x$ can range from $-\infty$ to $+\infty$
- The sigmoid (logistic) function squashes z into (0, 1)
- Output is interpreted as $P(Y = 1 \mid X)$
- A classification rule follows from a probability threshold

$$P(Y=1|X) = 1 / (1 + e^{-z}), \quad z = \beta_0 + \beta_1 x$$



Plot: The sigmoid curve mapping z to a probability

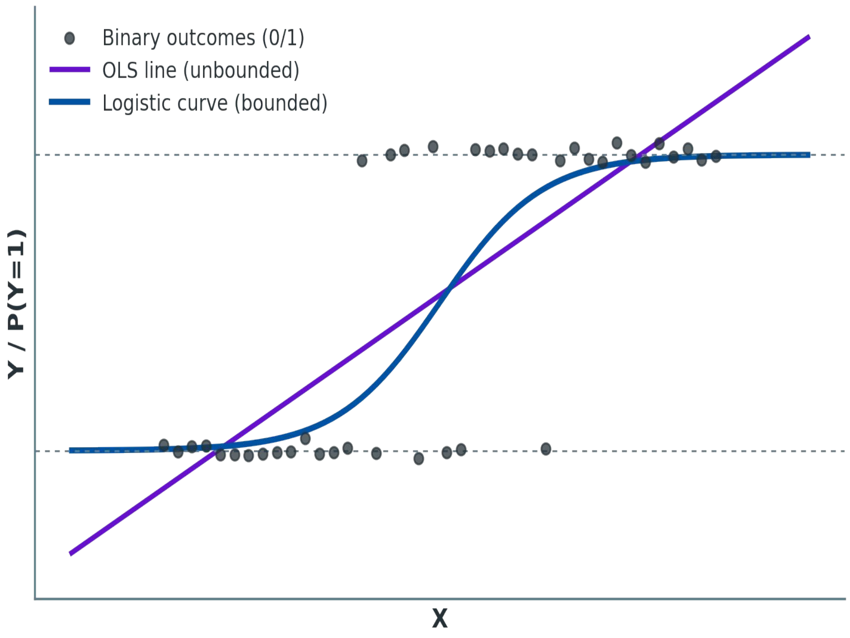
Log-Odds (Logit) Interpretation



- The logit is the log of the odds: ln(p / (1−p))
- The logit is linear in X — this is the actual "linear model" here
- Probability itself is a nonlinear, S-shaped function of X

Why OLS Doesn't Work for Binary Outcomes

- Y is 0/1, but an OLS line is unbounded — predicts below 0 or above 1
- Constant-variance and normality assumptions fail for binary Y
- Residuals are not normally distributed around a straight line
- A bounded, S-shaped curve is needed instead of a straight line



Plot: OLS line vs. logistic curve on binary outcome data

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Part 2

Estimation

Maximum Likelihood Estimation vs. OLS

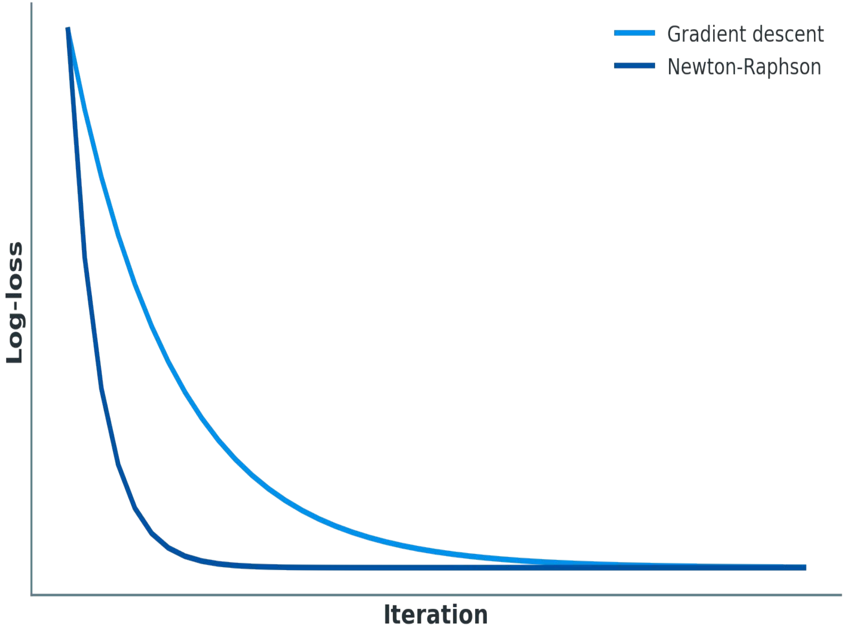
- OLS minimizes squared error — built for continuous, normal residuals
- Logistic regression instead maximizes the likelihood of the observed labels
- MLE finds β that makes the observed 0/1 outcomes most probable
- No closed-form solution exists — β must be found iteratively

Maximize: $L(\beta) = \prod p_i\text{^}\{y_i\}\ (1-p_i)\text{^}\{1-y_i\}$

Log-Loss and Iterative Optimization

- Log-loss (cross-entropy) is the negative log-likelihood, per observation
- Same optimization idea as linear regression: minimize a cost function
- Gradient descent: steepest-descent updates using $\nabla J(\beta)$
- Newton-Raphson: uses curvature (second derivatives) to converge faster

$$J(\beta) = -(1/n)\Sigma[y_i \ln(p_i)+(1-y_i)\ln(1-p_i)]$$



Plot: Log-loss vs. iteration, gradient descent vs. Newton-Raphson

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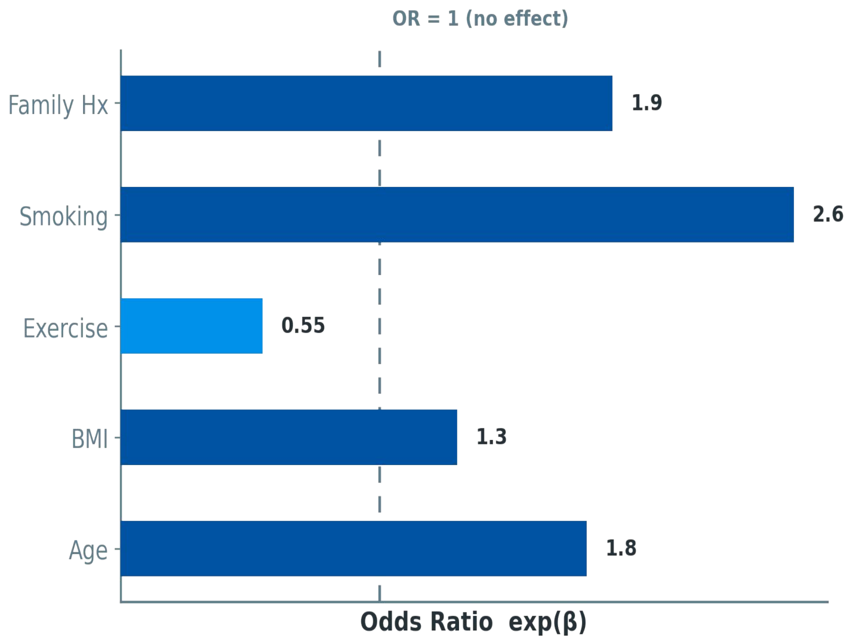
Part 3

Multivariable Logistic Regression

Coefficient Interpretation via Odds Ratios

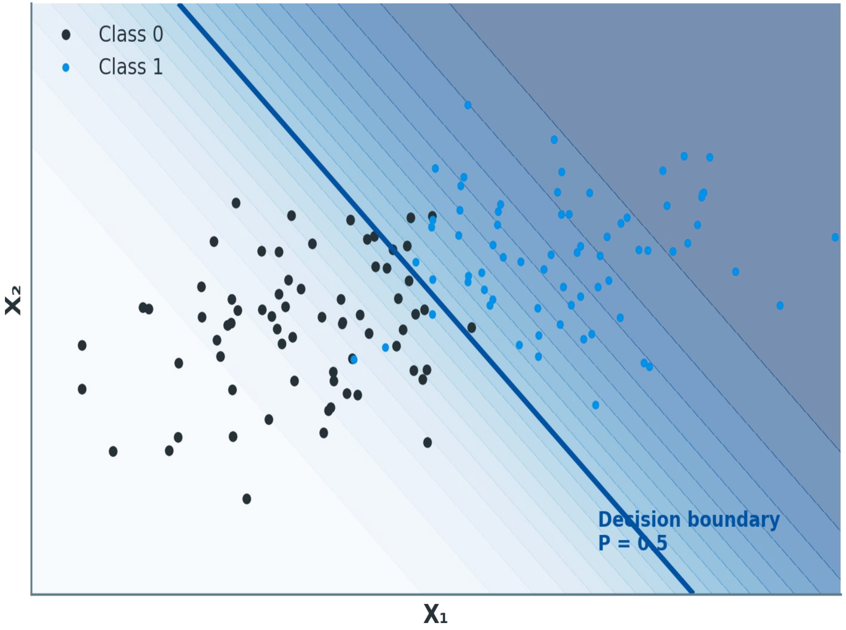
- exp(β□) is the odds ratio for a one-unit increase in X□
- OR > 1: predictor increases the odds of Y = 1
- OR < 1: predictor decreases the odds of Y = 1
- OR = 1: predictor has no effect on the odds



Plot: Odds ratios per predictor, referenced against OR = 1

Decision Boundary and Classification Threshold

- The decision boundary is where P(Y=1|X) = threshold
- Default threshold is 0.5 — predict class 1 if p ≥ 0.5
- With multiple predictors, the boundary is a hyperplane in logit space
- Raising or lowering the threshold trades precision for recall



Plot: Predicted probability contours and the P = 0.5 boundary

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Part 4

Key Metrics

The Confusion Matrix

	Predicted: 1	Predicted: 0
Actual: 1	**True Positive** Predicted 1, Actual 1	**False Positive** Predicted 1, Actual 0
Actual: 0	**False Negative** Predicted 0, Actual 1	**True Negative** Predicted 0, Actual 0

Accuracy, Precision, Recall, F1-Score

- Accuracy: overall share of correct predictions
- Precision: of predicted positives, how many are correct
- Recall (sensitivity): of actual positives, how many are found
- F1-score: harmonic mean of precision and recall

Accuracy

$(TP+TN) / (TP+FP+FN+TN)$

Precision

$TP / (TP+FP)$

Recall

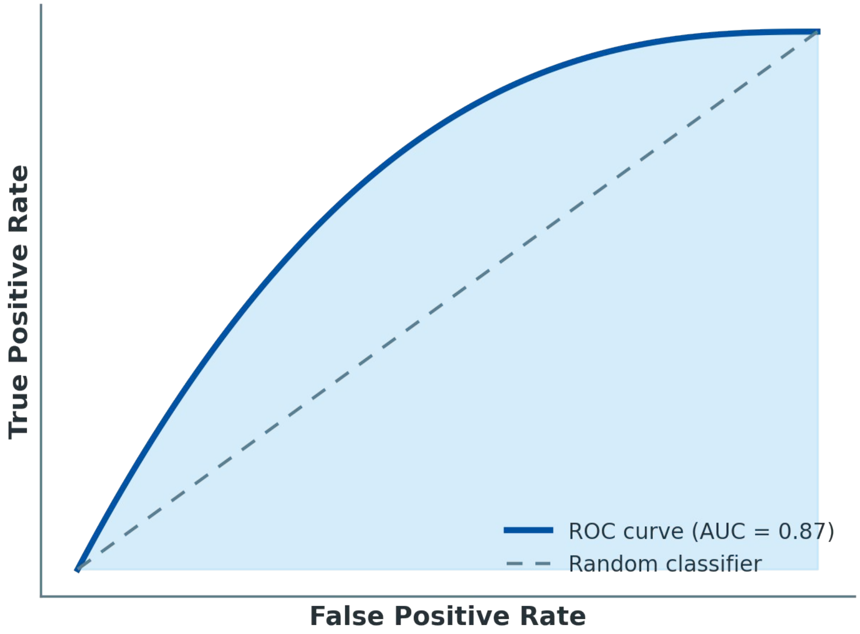
$TP / (TP+FN)$

F1-score

$2 \cdot (Prec \cdot Rec) / (Prec + Rec)$

ROC Curve and AUC

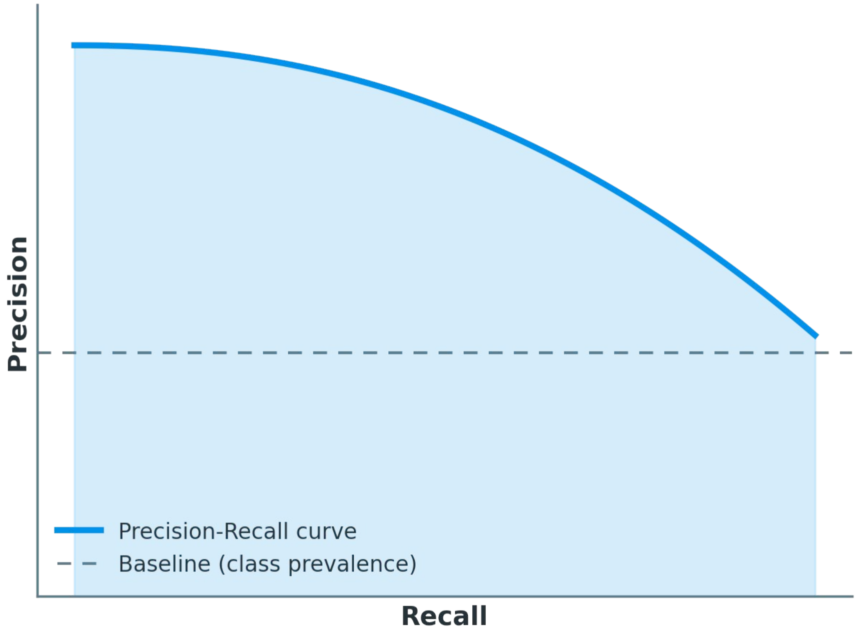
- ROC plots True Positive Rate vs. False Positive Rate
- Traces performance across every possible threshold
- AUC: probability the model ranks a random positive above a random negative
- AUC = 0.5 is random guessing; AUC = 1.0 is perfect separation



Plot: ROC curve with AUC shaded

Precision-Recall Curve and Log-Loss

- PR curve plots Precision vs. Recall across thresholds
- More informative than ROC when the positive class is rare
- Log-loss scores the model's actual probabilities, not just class labels
- Log-loss rewards confident, correct predictions and punishes confident errors



Plot: Precision-Recall curve vs. class-prevalence baseline

Choosing the Right Metric

Metric	Best used when…
Accuracy	Balanced classes, equal error cost
Precision	False positives are costly (e.g. spam filters)
Recall	False negatives are costly (e.g. disease screening)
F1-score	Need a single balance of precision and recall
ROC-AUC	Comparing rankings across thresholds, balanced classes
PR-AUC	Imbalanced classes, positive class is rare
Log-loss	Calibrated probabilities matter, not just the label

Thanks!

Any questions?